1 Introduction

In the exercises for Block 1 of the MSE module *TSM ModAvi*, you will apply the theoretical concepts explained in Videos 1 to 6 to a practical example. The exercises are divided into two parts: In **Exercise 1**, you will generate demand scenarios for an engineering system; in **Exercise 2**, you will determine (optimal) conventional and flexible facility requirements for that system.

2 Description of Practical Example

AEROVALE Regional Airport is a publicly owned regional airport located in a semi-rural area approximately 70 km from a major metropolitan hub. The airport primarily serves short-haul passenger traffic, regional cargo, and general aviation. It plays a key role in supporting regional economic development, tourism, and connectivity to national and European air transport networks.

The airport currently operates below its maximum physical capacity but faces strategic decisions related to future growth, environmental constraints, and operational efficiency.

Over the past decade, technological advances in electric propulsion, autonomy, and digital air traffic management have led to increased interest in Urban Air Mobility (UAM) and Uncrewed Aerial Vehicle (UAV) operations. Regional airports like *AEROVALE* are increasingly seen as potential nodes in future low-altitude air transport networks, rather than purely conventional aviation infrastructure.

The airport considers several non-exclusive future use cases:

- **Urban Air Mobility**: Electrical Vertical Take-off and Landing (eVTOL) aircraft, short-range passenger transport (10-100km), connections between the airport, nearby cities, and business districts.
- **UAV Operations:** Small- and medium-sized cargo drones, medical logistis (i.e., time-critical deliveries), infrastructure and surveillance.

These operations differ significantly from conventional aviation in terms of vehicle size, performance, noise characteristics, and infrastructure requirements.

3 Tasks

The overall objective of these exercises is to apply the methods taught in the videos of Block 1 to a practical example. Solve the tasks specified by writing, testing, and analysing software code (in Matlab, Python, or a coding language of your choice).

Below you will find a series of tasks that you must complete as part of this exercise. Please note that many of the tasks include a documentation requirement. You may integrate your

documentation directly into your code (e.g., as comments or as a markup cell in a Jupyter notebook) or document your work using a separate `.md` file.

Exercise 1:

1. **Select use cases to be considered in the strategic plan.** Identify and justify the use cases that you will consider for the development of airside facility requirements in the strategic plan of *AEROVALE* Regional Airport. Clearly specify which new or modified airside infrastructure is required to accommodate the new categories of airport users included in your use cases (e.g. UAVs, eVTOL aircraft).
   - *Note:* When determining facility requirements, **focus exclusively on the airside part** of *AEROVALE* Regional Airport (i.e., the stands, taxiways, runways, vertiports, etc.)
   - *Documentation:* Describe the selected use cases, new/modified airport users, required airside infrastrucure, and associated assumptions.
2. **Generate at least 1000 demand scenarios for your engineering system using Geometric Brownian Motion (GBM).** Estimate the GBM parameters from historical demand data for your selected system, following the method introduced in Video 2. Generate demand scenarios covering a planning horizon of at least 20 years.
   - *Hint:* To estimate realistic GBM parameters, use historical demand data of AEROVALE Regional Airport provided in document *aerovale_demand.csv*. This input data set covers the period between 1982 and 2024.
3. **Generate at least 1000 scenarios for your engineering system specifying the market share of UAVs, eVTOLs, etc.** In order to derive airside facility requirements for *AEROVALE* Regional Airport, you need, in addition to the demand scenarios (see Task 2), scenarios that describe when the new airport user groups included in your use cases (e.g. UAVs, eVTOL aircraft) enter the market and how large their market shares become over time. Use the same planning horizon as in Task 2.
   - *Note:* The generation of such scenarios is not explained in the videos or the lecture notes. Therefore, conduct a (brief) literature review and develop your own approach for generating scenarios that describe the market shares of these user groups.
   - *Documentation:* Describe the method you selected or developed to generate market share scenarios. Justify your method (i.e., explain why you consider this approach to be appropriate).
4. **Identify potential real options "in" and "on" systems** that could support the development of a flexible strategic plan for the airside infrastructure of *AEROVALE* Regional Airport.
   - *Documentation:* Describe at least two real options "in" systems and two real options "on" systems that could be applied.
5. **Define a candidate flexibility** based on your considerations in the previous task, which you intend to use to plan the system in a flexible manner.
   - *Hint:* Select a relatively simple candidate flexibility. You will later use this flexibility to derive flexible facility requirements based on decision rules.

- *Documentation:* Describe the selected candidate flexibility. Explain which real option(s) it is based on and how these option(s) operate.

Exercise 2:

6. **Collect information on the revenue and cost functions of your system.** Identify the operational and capital expenditures incurred by the system, their approximate magnitude, and the costs associated with capacity expansions. Identify the system's revenue streams and describe how they can be quantified. Define the revenue and cost functions you will later use to compute the expected net present value (ENPV) of your system, including the chosen functional forms and their parameterization.
   - *Hint:* See the *Airport Parking Garage Example* below for examples of revenue and cost functions that may inspire your own definitions.
   - *Documentation:* Describe your cost and revenue functions; list and define all parameters used (e.g., unit cost per unit of capacity); explain how you determined each parameter; and assess the reliability of your estimates.
7. **Determine the ENPV of your engineering system** over a planning horizon of 20 years (or more). Compute the ENPV for two planning strategies: (i) a conventional approach without real options, and (ii) a flexible approach based on the candidate flexibility defined in Task 4. In both cases, determine the facility requirements that maximize ENPV. To this end, use appropriate optimization algorithms or heuristics.
   - *Documentation:* Comment on your results. Do they meet your expectations? Are they realistic? How close to optimal do you consider your solution to be?
8. **Compare the optimal ENPV values** obtained for the conventional and flexible planning approaches. In addition, plot the target curves for both the conventionally planned and the flexibly planned system.
   - *Documentation:* Does the comparison align with your expectations? What conclusions and insights do you draw from comparing conventional and flexible planning?

4 Example: Airport Parking Garage

The following example of a parking garage at Zurich Airport shows how Exercises 1 and 2 could be implemented in practice. This parking garage example is loosley based on the paper of De Neufville et al. (2006).

This example illustrates how both conventional and flexible facility requirements can be formulated for a hypothetical parking garage at Zurich Airport. The strategic planning of this parking facility explicitly accounts for the anticipated increase in electrically powered vehicles in the future. In particular, the analysis examines whether it is economically beneficial for the airport to provide parking spaces equipped with charging infrastructure, allowing customers to charge their vehicles while parked.

To meet these planning assumptions, both conventional and flexible facility requirements are designed to account for uncertainties in (i) demand, represented by the number of required parking spaces, and (ii) the market share of electrically powered vehicles. The market share of electric vehicles is a key factor, as it determines the number of parking spaces that must

be equipped with charging infrastructure and influences the potential for the Flughafen Zürich AG to generate additional revenues through vehicle charging services.

4.1 Exercise 1: Generation of Scenarios

In order to derive facility requirements for the parking garage, two different types of scenarios are required. First, it is necessary to estimate how the demand for parking spaces will evolve in the future. Second, the future market share of electrically powered vehicles must be determined.

In a first step, we generate **parking space demand scenarios** for Zurich Airport analogue to the procedure documented in Video 2. In order to do so, let us make the following assumptions:

- Demand in year $t = 0$: $D_0 = 31195226$ annual passengers.
- Mean percentage growth rate $\mu_D = 4.96\%$, standard deviation of percentage growth rate $\sigma_D = 6.87\%$. We can estimate these two values using the methods presented in Video 2 from demand data published by Zurich Airport[1].
- Planning horizon $T = 20$ years

Applying these assumptions, we can determine passenger demand scenarios for Zurich Airport by using the methods introduced in Video 2. In a next step, these passenger demand scenarios must be converted into parking space demand scenarios. In order to to so, we are going to apply the following method (which is not covered in the videos, as it is airport sparking space specific):

1. According to Horonjeff et al. (2010)[2], an airport requires between 1000 to 1400 parking spaces per one million originating passenger.
2. Originating passengers are passengers which start their flight journey at the airport in question. If we subtract the amount of transfer passengers from the annual aggregated passengers (i.e., passengers who only transfer from one aircraft to another at the airport in question) and then divide the result by 2, we get the number of originating passengers. Example: Aggregated annual number of passengers: $D_0 = 31.19$ million, share of transfer passengers: 30%. This results in $\frac{31.19 \cdot 0.7}{2} = 10.91$ million originating passengers.
3. Calculate the required number of parking spaces by applying (1) on the number of passengers determined under (2). Example: For 10.91 million originating passengers, an airport requires $10.91 \cdot 1000 = 10910$ parking spaces.
4. In a final step, we have to consider how many of these parking spaces should be realized in the planned parking garage. Example: We assume that 20% of all parking spaces are to be realized in the parking garage. We therefore have a parking space requirement of $10910 \cdot 0.2 = 2184$ parking spaces in the parking garage of question.

By applying the conversion method introduced above to all passenger demand scenarios, we get parking space scenarios as, illustrated on the left in Figure 1.

[1] See https://newsroom.flughafen-zuerich.ch/statistisches-jahrbuch/

[2] Horonjeff, R., McKelvey, F. X., Sproule, W. J. & Young, S. B. (2010). *Planning and design of airports* (5th ed.). McGraw-Hill Education.

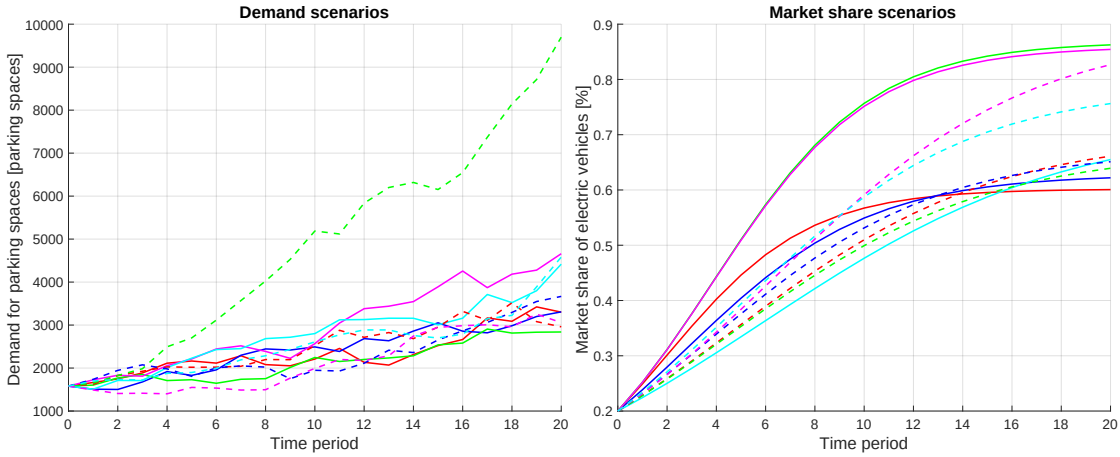

Figure 1: Demand scenarios generated for the parking garage (left) and market share scenarios for electric vehicles (right). Note, only 10 out of the 100 generated scenarios are shown in the figure.

In a second step, we define scenarios that specify the **market share of electrically powered vehicles** over time. To model this market share, we use an S-curve, i.e. a so-called *logistic diffusion model. Note: S-curves are not covered in the videos of Block 4.*

The S-curve assumes that the market share $S(t)$ of electrically powered vehicles at time t can be described by the following equation:

$$S(t) = \frac{S_T}{1 + \left(\frac{S_T - S_0}{S_0}\right) \cdot e^{-k \cdot t}} \tag{1}$$

Here, S_T denotes the long-run saturation level (i.e. the maximum market share reached by the end of the planning horizon), S_0 is the initial saturation level at time $t = 0$, and k is the growth speed. In 2024, the market share of electrically powered vehicles is approximately $S_0 = 20\,\%$. It is assumed that over a period of 20 years, the market share will increase to a level between $S_T = 60\,\%$ and $S_T = 90\,\%$. Reasonable values for the growth speed range from $k = 0.15$ (slow growth scenarios) to $k = 0.35$ (rapid growth scenarios).

The uncertainties associated with S_T and k can be incorporated into Equation 1 by assuming that both parameters are randomly distributed within the value ranges specified above. This leads to the following stochastic formulation of $S(t)$:

$$S(t) = \frac{S_T}{1 + \left(\frac{S_T - S_0}{S_0}\right) \cdot e^{-k \cdot t}} \quad \text{with } S_T \sim \mathcal{U}(0.6, 0.9) \text{ and } k \sim \mathcal{U}(0.15, 0.35) \tag{2}$$

where $\mathcal{U}$ denotes a uniformly distributed random variable. In *Microsoft Excel*, uniformly distributed random variables can be generated using the `RAND()` function. For example, the long-run saturation level S_T can be implemented as:

$$S_T = 0.6 + (0.9 - 0.6) \cdot \texttt{RAND()}$$

By applying Equation 2 to generate market share scenarios, we end up with S-curves as shown on the right in Figure 1.

4.2 Exercise 2: Cash Flow Estimation

This section illustrates how the cash flows generated by the parking garage can be estimated. To this end, we first have to define functions which specify (i) how much revenues the parking garage generates, and (ii) how much costs it incurs.

4.2.1 Revenue Function

Let us assume that the parking garage generates revenue through parking fees as well as through revenues generated from charging electric vehicles. To determine the annual revenues $\mathcal{R}_t$ generated by the parking garage in year t, we require the following information:

- r_P, the average parking fee revenues generated by a single parking space over the period of one year;
- r_E, the average annual revenues generated from charging electric vehicles;
- D_t, the demand for parking spaces in year t, i.e., demand scenarios, as determined in Section 4.1;
- S_t, the market share of electric vehicles in year t, i.e., market share scenarios, as determined in Section 4.1;
- K_t, the available / installed number of parking spaces in year t;
- $K_{t,E}$, the available / installed number of parking spaces allowing for electric charging in year t ($K_{t,E}$, is assumed to be 40 % of K_t); and
- p_E, the probability that an electric vehicle makes use of a charging facility.

To determine the annual revenue $\mathcal{R}_t$ generated by the parking garage, we first distinguish whether the demand for parking spaces D_t is less than or equal to the available parking capacity K_t, or whether it exceeds this capacity.

If $D_t \leq K_t$, the capacity of the parking garage is sufficient to serve all customers, and parking fees can be collected for all D_t vehicles. In this case, we further distinguish between two situations depending on the demand for electric charging. The demand for electric charging is given by $D_t \cdot S_t \cdot p_E$, where S_t denotes the market share of electric vehicles and p_E the probability that an electric vehicle makes use of a charging facility. If this charging demand does not exceed the available number of charging-capable parking spaces $K_{t,E}$, all charging demand can be served and charging revenues are generated for all $D_t \cdot S_t \cdot p_E$ vehicles. If, however, the charging demand exceeds $K_{t,E}$, charging revenues are limited by the available charging infrastructure, and revenues can only be generated for $K_{t,E}$ charging events.

If $D_t > K_t$, the demand for parking spaces exceeds the capacity of the parking garage. In this case, only K_t vehicles can be accommodated, while $D_t - K_t$ customers cannot be served. Consequently, parking fees can only be collected for K_t vehicles. As before, we distinguish whether the charging demand of the parked vehicles, given by $K_t \cdot S_t \cdot p_E$, is less than or equal to the available charging capacity $K_{t,E}$ or exceeds it. If the charging demand does not exceed $K_{t,E}$, charging revenues are generated for all $K_t \cdot S_t \cdot p_E$ vehicles. Otherwise, charging revenues are limited to $K_{t,E}$ charging events.

Based on these considerations, the annual revenue function $\mathcal{R}_t$ of the parking garage is given by Equation (3).

$$\mathcal{R}_t = \begin{cases} D_t \cdot r_P + D_t \cdot S_t \cdot p_E \cdot r_E & \text{if } D_t \leq K_t \text{ and } D_t \cdot S_t \cdot p_E \leq K_{t,E} \\ D_t \cdot r_P + K_{t,E} \cdot p_E \cdot r_E & \text{if } D_t \leq K_t \text{ and } D_t \cdot S_t \cdot p_E > K_{t,E} \\ K_t \cdot r_P + K_t \cdot S_t \cdot p_E \cdot r_E & \text{if } D_t > K_t \text{ and } K_t \cdot S_t \cdot p_E \leq K_{t,E} \\ K_t \cdot r_P + K_{t,E} \cdot S_t \cdot p_E \cdot r_E & \text{if } D_t > K_t \text{ and } K_t \cdot S_t \cdot p_E > K_{t,E} \end{cases} \tag{3}$$

4.2.2 Cost Functions

With regard to costs $\mathcal{C}_t$, we distinguish between capital expenses (CAPEX) $C_{CAPEX,t}$ and operational expenses (OPEX) $C_{OPEX,t}$. Capital expenses are incurred when the parking garage is initially built or expanded at a later point in time. Operational expenses result from daily business activities. If we assume that the parking garage is built from scratch in year $t = 0$, the cost function $\mathcal{C}_t$ can be written as

$$\mathcal{C}_t = \begin{cases} C_{CAPEX,0} & \text{if } t = 0 \\ C_{CAPEX,t} + C_{OPEX,t} & \text{if } t > 0. \end{cases} \tag{4}$$

Capital Expenses. $C_{CAPEX,t}$ are incurred either when the parking garage is built initially or when its capacity is changed by $\Delta K_t = K_t - K_{t-1}$, which in simple terms means that ΔK_t new parking spaces are added somehow (e.g., by adding another floor to the building or by changing the footprint of the building). Irrespective of whether we consider the initial construction of the facility or an expansion, the CAPEX resulting of capacity change by ΔK_t can be estimated as

$$C_{CAPEX,t} = (K_t - K_{t-1})^{\alpha} \cdot c_{\text{CAPEX}} = (\Delta K_t)^{\alpha} \cdot c_{\text{CAPEX}} \tag{5}$$

where parameter α refers to the so-called economies of scale factor, and c_{CAPEX} specifies the unit costs of installation, which is the price of one single parking space in our case.

In the parking garage, we must further distinguish between *conventional* parking spaces (i.e., parking spaces without the possibility to charge electric vehicles) and *electric* parking spaces that allow for vehicle charging. Since the installation of charging infrastructure incurs additional costs, this distinction must be explicitly taken into account when calculating the capital expenditures (CAPEX).

$$C_{CAPEX,t} = (\Delta(K_t - K_{t,E}))^{\alpha} \cdot c_{\text{CAPEX,conv}} + (\Delta K_{t,E})^{\alpha} \cdot c_{\text{CAPEX,elec}} \tag{6}$$

Operational Expenses. $C_{OPEX,t}$ include costs for leasing the land on which the parking garage is located as well as costs arising from maintenance and operation of the facility. As with the revenues, we must distinguish between the cases $D_t \leq K_t$ and $D > K_t$. Thus, OPEX can be formulated as follows

$$C_{OPEX,t} = \begin{cases} D_t \cdot c_{ops} + c_{\text{lease}} & \text{if } D_t \leq K_t \\ K_t \cdot c_{ops} + c_{\text{lease}} & \text{if } D_t > K_t \end{cases} \tag{7}$$

where c_{ops} are operating costs incurred per parking space over one planning period t and c_{lease} are the leasing costs for the land the parking garage is build upon for one planning period t.

To keep the formula for $C_{OPEX,t}$ as simple as possible, we assume that there are no additional operational costs for parking spaces with electric chargers.

4.2.3 Net Present Value

Since revenues $\mathcal{R}_t$ and costs $\mathcal{C}_t$ depend on demand and the market share of electric vehicles, the net present value (NPV) is to be calculated for every scenario you generated as outlined in Section 4.1. For a given demand and market share scenario, you first calculate costs and revenues for every planning period $t = 1, \ldots, T$ considered. Then, you derive the profit $\mathcal{R}_t - \mathcal{C}_t$ in year t in order to determine the NPV_s of your system for the demand scenario s as

$$NPV_s = -\mathcal{C}_0 + \sum_{t=1}^{T} \frac{1}{(1+\delta)^t} \cdot (\mathcal{R}_t - \mathcal{C}_t) \tag{8}$$

where δ is the discount factor. After repeating this procedure for each and every scenario, you can determine the expected net present value (ENPV) of your system as

$$ENPV = \frac{1}{S} \cdot \sum_{i=1}^{S} NPV_s \tag{9}$$

where S is the number of scenarios considered, and NPV_s is the NPV resulting from scenario $s = 1, 2, \ldots, S$. Indeed, Equation 9 is nothing but the arithmetic mean[3] over all NPV_s.

4.3 Exercise 2: Conventional Facility Requirements

Conventional facility requirements can be imagined as schedules: They describe how much capacity K_t (in our case, parking spaces) is made available in the parking garage at time t. Given conventional facility requirements as well as demand and market share scenarios, see Section 4.1, revenues generated and costs incurred by the system can then be determined. This way, the NPV_s of the parking garage for each demand scenario s, and ultimately the ENPV of the system across all demand scenarios $s = 1, 2, \ldots, S$ can be derived as mentioned in Equations 8 and 9.

4.4 Exercise 2: Flexible Facility Requirements

When defining flexible facility requirements, we assume that the parking garage is built with a real option *in* system, which allows the addition of building levels at a later point in time. To model this real option *in* system, we use the following *conditional-go* decision rule:

- **IF** $K_{t-1} - D_t < \theta_1$
- **THEN** $K_t = K_{t-1} + \theta_2$
- **ELSE** $K_t = K_{t-1}$

[3] In Microsoft Excel, you can calculate the arithmetic mean with *=MITTELWERT()* or *=MEAN()*, depending on your language settings.

This decision rule states that the capacity of the parking garage is increased by an amount of θ_2 parking spaces if condition $K_{t-1} - D_t < \theta_1$ is true. In this context, it is important to note that the decision rule is executed at every planning time t. At this point in time, the current demand D_t and the capacity K_{t-1} installed at the previous planning period $t-1$ are taken into account in the decision rule. Figure 2 shows graphically how θ_1 is calculated and how θ_2 affects capacity K_t.

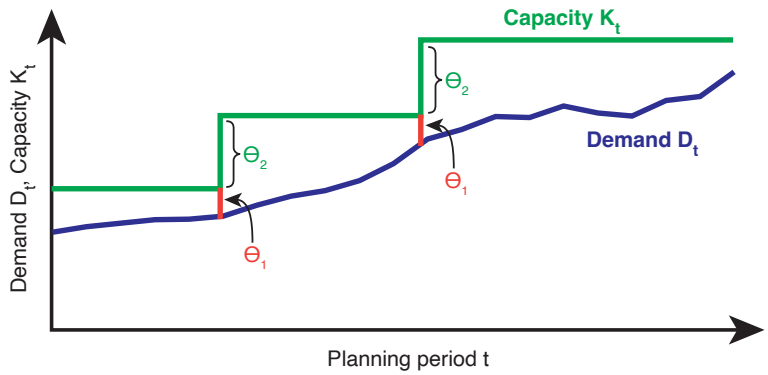

Figure 2: Functioning of the exemplary decision rule. The capacity of the engineering system is adjusted by θ_2 whenever $D_t - (K_{t-1} \cdot TH) > \theta_1$ applies.

4.5 Exercise 2: Determination of Optimal Facility Requirements

Solution Technique. Facility requirements are considered optimal if they maximize the ENPV of an engineering system. In practice, this means that for conventional facility requirements you have to find the schedule that maximizes the ENPV, while for flexible facility requirements you have to find the values for θ_1 and θ_2 that maximize the ENPV. However, this is easier said than done: There is a huge number of potentially feasible schedules (conventional case) and values for θ_1 and θ_2 [4]. To find optimal facility requirements, powerful solution methods are documented in the literature. However, these techniques are beyond the scope of this course. Subsequently, in PPP we apply the most simple (and the least performing) method to find optimal solutions: We define possible solutions by hand and try to find the best possible solution by cleverly adapting and adjusting these solutions. To find optimal conventional facility requirements, three factors can be *"played"* with: (i) the initial capacity of the system K_0, (ii) the point in times when the capacity of the system is expanded, and (iii) by how much the system is expanded in each case. There are also three adjusting screws for flexible facility requirements: (i) the initial capacity of the system K_0, (ii) the value of θ_1, and (iii) the value of θ_2.

Setting up a Baseline. In the following, I will show you how to find a good (but not optimal) solution for both the conventional as well as the flexible problem. Before we can start this search, we need to agree on a *baseline*. This refers to a set of default parameters from which we start our search. For the parking garage example, I have used the following default values:

- Average revenues from parking fees per parking space and year, $r_P = 15000$ CHF.

[4] Especially in the case of conventional facility requirements there can be millions or even billions of potential solutions amongst which one has to find the optimal one. Subsequently, the search for the optimal solution is nothing but trivial.

- Average revenues from electrical charging fees per parking space and year, $r_E = 4000$ CHF.
- Lease of land $c_{\text{lease}} = 2000000$ CHF for the entire parking garage per planning phase t
- Annual operating costs per parking space $c_{ops} = 3000$ CHF
- Construction costs per *conventional* parking space $c_{CAPEX,conv} = 60000$ CHF, and for parking space allowing *charging* $c_{CAPEX,elec} = 1.3 \cdot c_{CAPEX,conv}$
- Percentage of users that charge car, $p_E = 20\%$
- Discount factor $\delta = 4\%$
- Parking spaces per building level: 400
- Initial number of parking spaces: $K_0 = 5$ levels $\cdot$ 400 parking spaces $= 2000$ parking spaces
- We assume that 40 % percent of all parking spaces are (to be) equipped with electric chargers.

The basic idea of searching an optimal solution is rather simple: We only change one variable at a time and check at how the ENPV of the system changes. In the conventional case, we can change the initial size of the system K_0, as well as the number and amount of capacity adjustments in the period $t = 1, 2, \ldots, T$, while in the flexible case we can vary K_0, θ_1, and θ_2.

Finding Optimal Initial Building Size. Let us start with determining the optimal initial size K_0^* of the parking garage. Figure 3 illustrates how the initial capacity K_0 of the car park, specified in the number of building levels (each one containing 400 parking spaces), influences the ENPV of the system. The red line describes the ENPV of a conventionally planned system, while the blue line illustrates the ENPV of a flexibly planned system.

Figure 3 enables us to draw the following conclusions:

- For a value of $K_0 \leq 5$ building levels, the ENPV of a conventionally planned parking garage is lower than the one of a flexibly planned system. The opposite, however way less pronounced, it true for $K_0 > 5$ building levels.
- For $K_0 = 5$ building levels, the ENPV of the system, irrespectively of planned flexibly or conventionally, is highest. For this reason, we assume that $K_0^* = 5$ building levels.

Finding Optimal Parameters for Conventional Facility Requirements. Having determined the optimal initial size of the parking garage K_0^*, we can now focus on how the garage capacity should be adjusted over the planning horizon. For conventional facility requirements, we assume that the capacity of the parking garage is adjusted only once during the time interval between $t = 1$ and $t = T$. The time at which this adjustment occurs is denoted by $t_{\Delta K}$, while the magnitude of the adjustment (i.e. the number of parking spaces added at time $t_{\Delta K}$) is denoted by ΔK.

The illustration on the left-hand side of Figure 4 shows the expected net present value (ENPV) of the parking garage as a function of the timing of the capacity adjustment $t_{\Delta K}$ for five different adjustment sizes, $\Delta K = 400, 800, \ldots, 2000$ parking spaces. For each curve, the maximum ENPV is highlighted with a dot. This representation allows

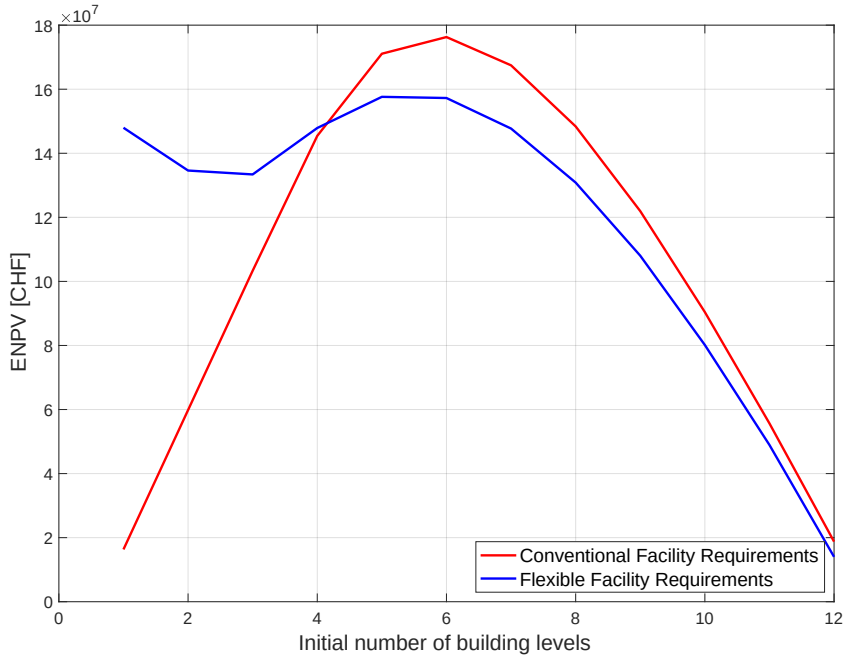

Figure 3: Influence of initial building size K_0 on the ENPV of the parking garage.

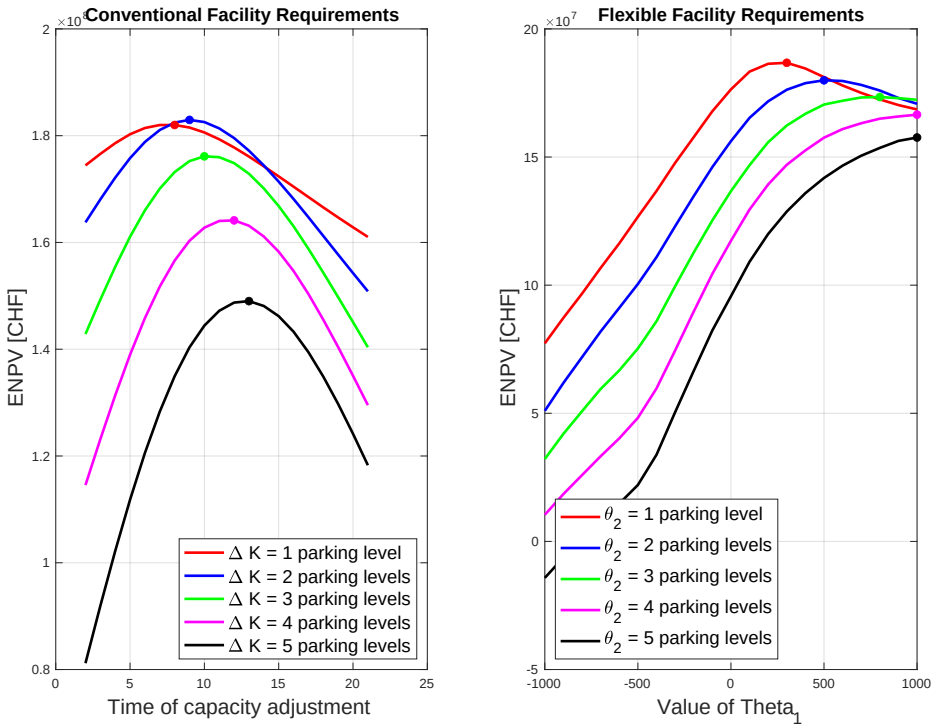

Figure 4: Influence of capacity adjustment time $t_{\Delta K}$ and size ΔK in the conventional case (left side), as well as θ_1 and θ_2 values in the flexible case (right side) on ENPV of parking garage.

us to draw the following conclusion: for a conventionally planned system, the ENPV is highest when the capacity adjustment is relatively *small* (approximately 800 parking spaces) and is implemented *early* in the planning horizon (around $t = 9$). Consequently, we use $t^*_{\Delta K} = 9$ years and $\Delta K^* = 2$ parking levels as optimal values for conventional facility requirements.

Finding Optimal Parameters for Flexible Facility Requirements. Analogous considerations can be made for a flexibly planned parking garage. The right-hand plot in Figure 4 illustrates how the choice of the decision-rule parameters θ_1 and θ_2 influences the ENPV of the system. The parameter θ_1 (i.e., the difference between D_t and K_{t-1} required to trigger a capacity expansion) is shown on the x-axis, while different values of θ_2 are represented by curves of different colors.

As in the case of conventional facility requirements, it can be observed that the ENPV decreases as the size of the capacity adjustment θ_2 increases. The results in Figure 4 indicate that the optimal value of θ_1 is approximately 300 parking spaces, while the optimal value of θ_2 is around 400 parking spaces (which is equal to 1 building level). Consequently, we use $\theta^*_1 = 300$ parking spaces and $\theta^*_2 = 1$ parking level.

Comparison Conventional vs. Flexible Planning. Finally, we compare the financial performance of a conventionally planned system with that of a flexibly planned system. To this end, we refer to Figure 5, where the solid lines represent so-called *target curves*, i.e. the empirical cumulative distribution functions (ECDFs) of the net present value outcomes NPV_s across all simulated scenarios. The conventionally planned system is shown in red, while the flexibly planned system is shown in blue.

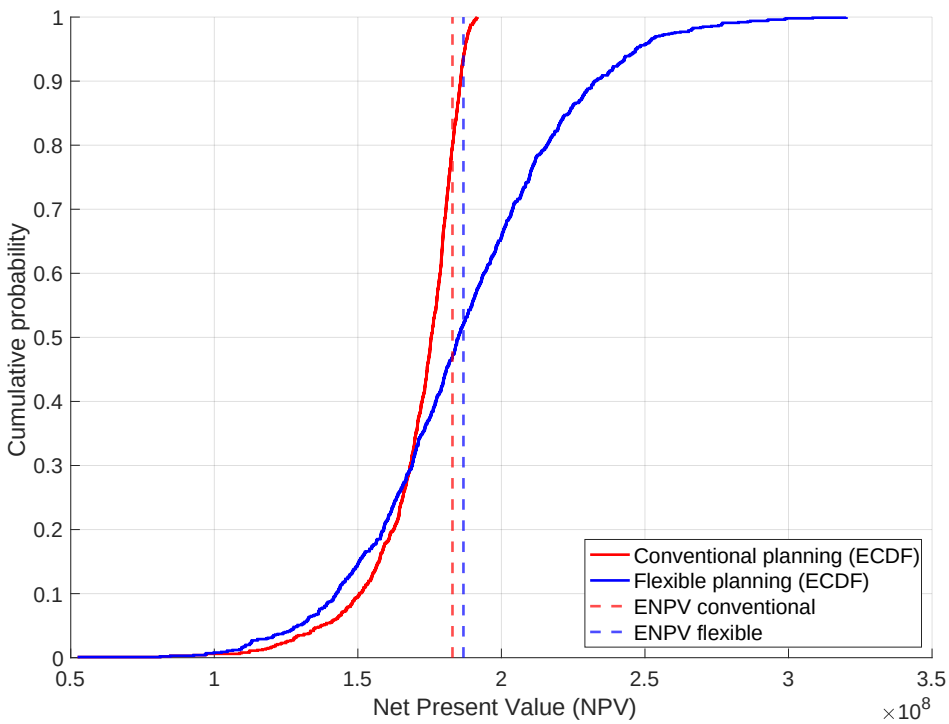

Figure 5: Target curves and expected net present values (ENPV) of the conventionally and flexibly planned parking garage

As can be seen from the figure, the flexibly planned parking garage performs substantially better in capitalizing on upside opportunities when future conditions turn out to be favourable. This is evidenced by the more pronounced right-hand tail of the blue target curve compared to the red curve, indicating a higher probability of achieving large positive NPVs.

In addition, a comparison of the expected net present values (ENPV), indicated by the dashed vertical lines in the figure, shows that the flexibly planned system achieves an ENPV of 188.7×10^6 CHF, whereas the conventionally planned system results in a lower ENPV of 184.6×10^6 CHF.

List of References

De Neufville, R., Scholtes, S., & Wang, T. (2006). Real options by spreadsheet: Parking garage case example. *Journal of Infrastructure Systems*, *12*(2), 107–111.